

→ The pumping lemma for Context-Free language is a theoretical property that every context-free language must satisfy.

- It is mainly used to prove that a language is not context-free.

- It is related to:

- Context-Free Grammars (CFG)
- Pushdown Automaton (PDA)

(can be written as ::

$$w = uvxyz$$

such that:

i) $|vxy| \leq p$

ii) $|vy| \geq 1$

iii) For all $i \geq 0$:

$$u v^i x y^i z \in L$$

$u, v, x, y, z \rightarrow$ Substrings of w

$p \rightarrow$ Pumping length

v and $y \rightarrow$ parts that can be repeated

① Application:

• ~~understand hierarchy:~~

• ~~Regular & Context~~

- Proving a language is not context-free
- Analyze grammar power
- Understand automaton limitations
- Prove theoretical properties